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[22BS2MA03]

SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2023

DISCRETE MATHEMATICS AND GRAPH THEORY

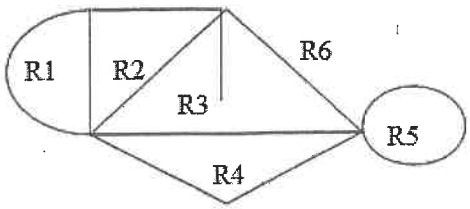
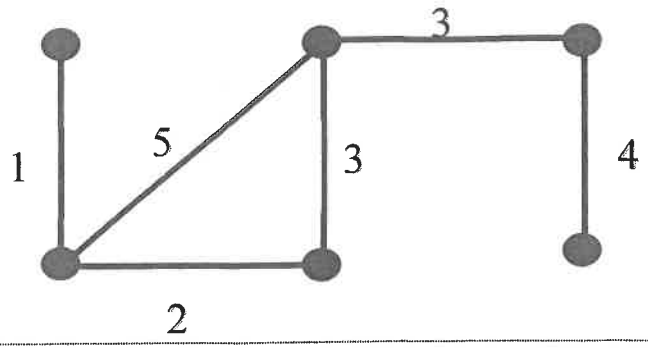
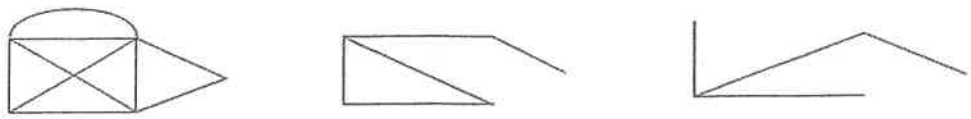
Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

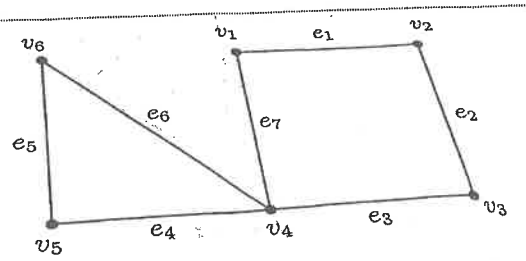
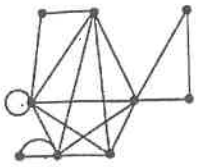
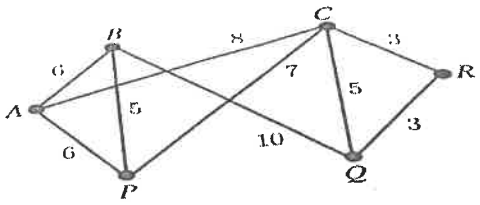
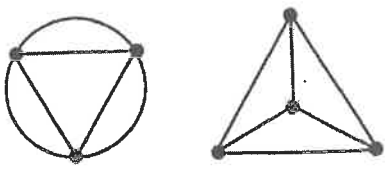
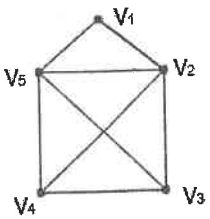
ii. Answer All Questions

PART-A **05 X 02=10**

	a.	How many different ways a shirt and a tie can be chosen, if a person has 8 shirts and 5 ties.	L-2	CO2	**02**
	b.	Find the generating function for the sequence 1, 1, 1, 1.....	L-2	CO1	**02**
	c.	Solve the recurrence relation $a_{n+1} = 4a_n$, given that $a_0 = 3$	L-3	CO1	**02**
	d.	Find the Chromatic number of Kuratowski first graph.	L-2	CO1	**02**
	e.	Explain complete binary tree.	L-2	CO1	**02**
		PART-B **06 X05=30**			
2.	a.	Find the number of positive integer solutions of the equation $x_1 + x_2 + x_3 = 17$ we have $x_1 \geq 1,\ x_2 \geq 1,\ x_3 \geq 1$.	L-3	CO3	**05**
		OR			
	b.	How many friends must you have to guarantee that at least 5 of them will have birthdays in the same month?	L-3	CO3	**05**
3.	a.	Determine the coefficient of x^{18} in th expansion of $(x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + \cdots)^5$	L-3	CO4	**05**
		OR			
	b.	Find the number of ways of forming a committee of 9 students drawn from 3 different classes in such a way that students from the same class do not have an absolute majority in the committee.	L-3	CO4	**05**
4.	a.	The number of viruses affected files in a system is 1000 (to start with) and this increases 250% every two hours. Use a recurrence relation to determine the number of viruses affected files in the system after one day.	L-3	CO2	**05**
		OR			
	b.	Solve the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$.	L-3	CO4	**05**

5.	**a.**	Define the following with an example for each. **i.** Graph **ii.** Simple Graph **iii.** Regular Graph	L-2	CO5	**05**
		OR			
	b.	Verify Euler's formula for the planar graph. Find the degree of each Region and show that sum of degree of regions is equal to twice the number of edges.  	L-3	CO5	**05**
6.	**a.**	Using the merge sorting method sort the following list -1, 0, 2, -2, 3, 6, -3, 5, 1, 4	L-3	CO5	**05**
		OR			
	b.	Find all possible spanning trees of the graph given below and also identify minimal spanning tree.  	L-3	CO5	**05**
7.	**a.**	Define the following with an example for each **i.** Euler Trail & Euler Circuit **ii.** Hamilton Path & Hamilton Cycle	L-2	CO5	**05**
		OR			
	b.	For the graph G and its subgraph G1 and G2 find the complement of its subgraph.  G G1 G2	L-3	CO5	**05**

		PART-C **06 X 10=60**			
8.	**a.**	In how many ways can 26 letters of English alphabet be permuted so that none of the pattern's CAR, DOG, PUN or BYTE occurs.	L-3	CO2	**10**
		OR			
	b.	Five teachers T1, T2, T3, T4, T5 are to be made class teachers for five class C1, C2, C3, C4, C5, one teacher for each class. T1 and T2 do not wish to become the class teachers for C1 or C2, T3 and T4 for C4 or C5, and T5 for C3 or C4 or C5. In how many ways can the teachers be assigned work (without displeasing any teacher)?	L-3	CO3	**10**
9.	**a.**	Using generating function, find the number of partitions of n = 6.	L-3	CO4	**10**
		OR			
	b.	For each of the words given below, find the exponential generating function for the number of ways to arrange $n \geq 11$ letters selected from the word. **i.** MISSISSIPPI **ii.** ISOMORPHISM **iii.** MATHEMATICS **iv.** ENGINEERING **v.** ENGINE	L-3	CO4	**10**
10.	**a.**	Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with the initial conditions $a_0 = 1$ and $a_1 = 4$.	L-3	CO4	**10**
		OR			
	b.	Find the generating function for the recurrence relation $a_{n+1} - a_n = 3^n, \quad n \geq 0,$ with $a_0 = 1$ and hence solve the relation.	L-3	CO4	**10**
11.	**a.**	For the graph given below, indicate the nature of the following walks. **i.** $v_1 e_1\ v_2\ e_2\ v_3\ e_2\ v_2$ **ii.** $v_4 e_7\ v_1\ e_1\ v_2\ e_2\ v_3\ e_3\ v_4\ e_4\ v_5$ **iii.** $v_1 e_1\ v_2\ e_2\ v_3\ e_3 v_4\ e_4\ v_5$ **iv.** $v_1 e_1\ v_2\ e_2\ v_3\ e_3 v_4 e_7 v_1$ **v.** $v_6 e_5\ v_5\ e_4\ v_4\ e_3\ v_3 e_2 v_2\ e_1\ v_1\ e_7 v_4 e_6 v_6$	L-3	CO5	**10**

					
		OR			
	b.	Check the planarity (or otherwise) of the following graph by the method of elementary reduction 	L-3	CO5	**10**
12.	**a.**	Obtain an optimal prefix code for the message ROAD IS GOOD. Also Indicate the code.	L-3	CO5	**10**
		OR			
	b.	Using Kruskal's Algorithm, find a minimal spanning tree for the given weighted graph. 	L-3	CO5	**10**
13.	**a.**	Construct the Dual of the following planar graph.  (a) (b)	L-3	CO5	**10**
		OR			
	b.	Find the chromatic polynomial for the graph shown below using multiplication theorem. If 5 colors are available, in how many ways can the vertices of this graph be properly colored? 	L-3	CO5	**10**